Fuzzy Relations :

Let R be the relation defined on two fuzzy sets say A and B on the universe of discourse X & Y respectively with

$\mu_A(x)/x \in X$ & $\mu_B(x)/y \in Y$,

then

$R = A \times B$, where R has a membership function given by

$$\mu_{A \times B}(x, y) = \min \{(\mu_A(x), \mu_B(y)\}$$

Eg. Let $A = \left\{\frac{0.1}{X_1}, \frac{0.4}{X_2}, \frac{0.7}{X_3}\right\}$ & $B = \left\{\frac{0.2}{Y_1}, \frac{0.4}{Y_2}, \frac{0.9}{Y_3}\right\}$

Let R be the relation on A & B, then

R =	Y_1	Y_2	Y_3
X_1	0.1	0.1	0.1
X_2	0.2	0.4	0.4
X_3	0.2	0.4	0.7

Compositions of fuzzy Relation :

Let R & S be two relations defined on the set X to Y & Y to Z respectively given by

$$R = \begin{array}{c} \\ X_1 \\ X_2 \end{array} \begin{array}{c} \begin{array}{cc} Y_1 & Y_2 \end{array} \\ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \end{array} \quad \& \quad S = \begin{array}{c} \\ Y_1 \\ Y_2 \end{array} \begin{array}{c} \begin{array}{cc} Z_1 & Z_2 \end{array} \\ \begin{bmatrix} x & y \\ z & w \end{bmatrix} \end{array}$$

then composition of R & S is denoted by R∘S from X to Z & is given by

$$R \circ S(X_1, Z_1) = \text{Max}\left[\min(a,x), \min(b,z)\right]$$

$$R \circ S(X_1, Z_2) = \text{Max}\left[\min(a,y), \min(b,w)\right]$$

$$R \circ S(X_2, Z_1) = \text{Max}\left[\min(c,x), \min(d,z)\right]$$

$$R \circ S(X_2, Z_2) = \text{Max}\left[\min(c,y), \min(d,w)\right]$$

- Union of fuzzy relations :

Let R and S be two fuzzy relations defined on the A×B , then

$$\mu_{R \cup S}(a,b) = \text{Max}\left\{\mu_R(a,b), \mu_S(a,b)\right\}$$

- Intersection of Fuzzy Relations :

Let R and S be two fuzzy relations defined on the A×B, then

$$\mu_{R \cap S}(a,b) = \min\{\mu_R(a,b), \mu_S(a,b)\}$$

- Complement of Fuzzy Relations :

Let R be the relations defined on the set A×B, then complement of R is denoted by $\overline{R}$ or R^c & is given by

$$\mu_{\overline{R}}(a) = \mu_{R^c}(a) = 1 - \mu_R(a)$$

Note: The union & intersection of fuzzy relation exists only if they are defined on same universe of discourse.

Q1. If Fuzzy Sets $A = \left\{\frac{0.3}{30} + \frac{0.7}{60} + \frac{1}{100}\right\}$,

$B = \left\{\frac{0.2}{20} + \frac{0.4}{40} + \frac{0.6}{60}\right\}$ & $C = \left\{\frac{0.3}{50} + \frac{0.6}{100}\right\}$

Find Fuzzy relation $R = A \times B$, $S = B \times C$.

Sol? We have $R = A \times B$ &

$$\mu_{A\times B}(x,y) = \mu_R(x,y) = \min\{\mu_A(x), \mu_B(y)\}$$

$$R = A\times B = \left\{\frac{0.2}{(30,20)} + \frac{0.3}{(30,40)} + \frac{0.3}{(30,60)} + \frac{0.2}{(60,20)} + \frac{0.4}{(60,40)} + \frac{0.6}{(60,60)} + \frac{0.2}{(100,20)} + \frac{0.4}{(100,40)} + \frac{0.6}{(100,60}\right\}.$$

In matrix form

	20	40	60
30	0.2	0.3	0.3
R = 60	0.2	0.4	0.6
100	0.2	0.4	0.6

Now

$$S(x,y) = (B\times C)(x,y)$$

We have

$$\mu_S(x,y) = \mu_{(B\times C)}(x,y) = \min\{\mu_B(x), \mu_C(y)\}$$

$S = B \times C$

$$= \left\{\frac{0.2}{(20,50)} + \frac{0.2}{(20,100)} + \frac{0.3}{(40,50)} + \frac{0.4}{(40,100)} + \frac{0.3}{(60,50)} + \frac{0.6}{(60,100)}\right\}$$

In matrix form

$$S = B \times C = \begin{array}{c} \\ 20 \\ 40 \\ 60 \end{array} \begin{array}{c} \begin{array}{cc} 50 & 100 \end{array} \\ \begin{bmatrix} 0.2 & 0.2 \\ 0.3 & 0.4 \\ 0.3 & 0.6 \end{bmatrix} \end{array}$$

Ex.23. Let $A = \{a_1, a_2, a_3\}$ and $B = \{b_1, b_2, b_3, b_4\}$.

Let R be the relation from A to B given by $\frac{0.1}{(a_1, b_3)} + \frac{0.8}{(a_1, b_4)}$

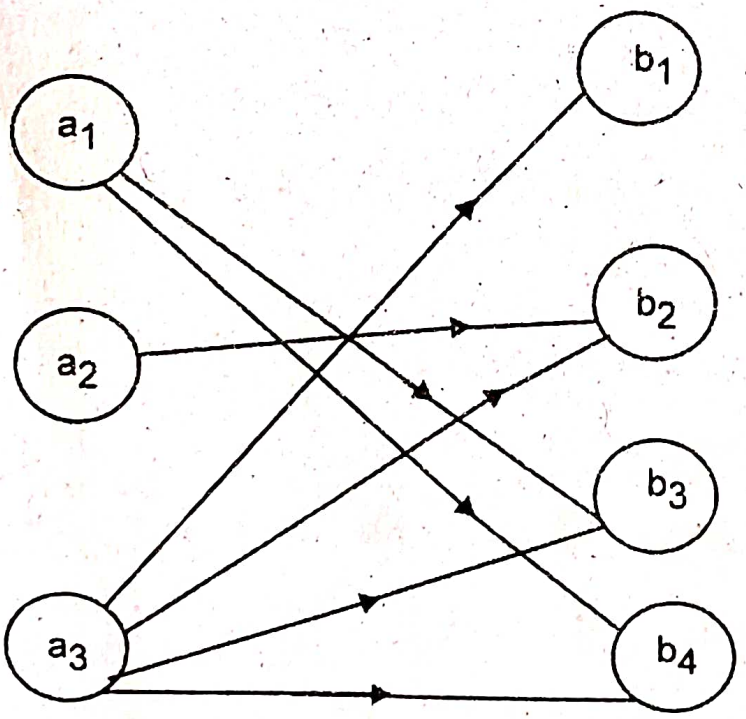
$$+\frac{0.8}{(a_2, b_2)} + \frac{0.1}{(a_3, b_1)} + \frac{0.8}{(a_3, b_2)} + \frac{1}{(a_3, b_3)} + \frac{0.8}{(a_3, b_4)}$$

Write fuzzy matrix M_R and draw graph.

Soln. We know that, the relation R from A to B in matrix form is denoted by M_R and is known as fuzzy matrix which can be written as follows:

$$M_R = \begin{array}{c} \\ a_1 \\ a_2 \\ a_3 \end{array} \begin{array}{c} \begin{array}{cccc} b_1 & b_2 & b_3 & b_4 \end{array} \\ \begin{bmatrix} 0 & 0 & 0.1 & 0.8 \\ 0 & 0.8 & 0 & 0 \\ 0.1 & 0.8 & 1 & 0.8 \end{bmatrix} \end{array}$$

Graph: In graphical form, the above relation can be shown as



Ex.24. If $A = \left\{\frac{0.3}{30}, \frac{0.7}{60}, \frac{1}{100}, \frac{0.2}{120}\right\}$ and $B = \left\{\frac{0.2}{20}, \frac{0.4}{40}, \frac{0.6}{60}, \frac{0.8}{80}\right\}$

are fuzzy sets, then find the relation $R = A \times B$. Also find R^2.

Soln. We have, $R = A \times B$ and

$$\Rightarrow \mu_{A\times B}(x, y) = \min\{\mu_A(x), \mu_B(y)\}$$

$$\therefore \quad R = \left\{\frac{0.2}{(30, 20)}, \frac{0.3}{(30, 40)}, \frac{0.3}{(30, 60)}, \frac{0.3}{(30, 80)}, \frac{0.2}{(60, 20)}, \frac{0.4}{(60, 40)}, \right.$$

$$\frac{0.6}{(60, 60)}, \frac{0.7}{(60, 80)}, \frac{0.2}{(100, 20)}, \frac{0.4}{(100, 40)}, \frac{0.6}{(100, 60)}, \frac{0.8}{(100, 80)},$$

$$\left.\frac{0.2}{(120,20)}, \frac{0.2}{(120,40)}, \frac{0.2}{(120,60)}, \frac{0.2}{(120,80)}\right\}$$

Now, In matrix form

$$R = M_R = \begin{array}{c} \\ 30 \\ 60 \\ 100 \\ 120 \end{array} \begin{array}{c} \begin{array}{cccc} 20 & 40 & 60 & 80 \end{array} \\ \begin{bmatrix} 0.2 & 0.3 & 0.3 & 0.3 \\ 0.2 & 0.4 & 0.6 & 0.7 \\ 0.2 & 0.4 & 0.6 & 0.8 \\ 0.2 & 0.2 & 0.2 & 0.2 \end{bmatrix} \end{array}$$

Now

$$R^2 = R \circ R = \begin{array}{c} \\ 30 \\ 60 \\ 100 \\ 120 \end{array} \begin{array}{c} \begin{array}{cccc} 20 & 40 & 60 & 80 \end{array} \\ \begin{bmatrix} 0.2 & 0.3 & 0.3 & 0.3 \\ 0.2 & 0.4 & 0.6 & 0.7 \\ 0.2 & 0.4 & 0.6 & 0.8 \\ 0.2 & 0.2 & 0.2 & 0.2 \end{bmatrix} \end{array}$$

$$\circ \begin{array}{c} \\ 30 \\ 60 \\ 100 \\ 120 \end{array} \begin{array}{c} \begin{array}{cccc} 20 & 40 & 60 & 80 \end{array} \\ \begin{bmatrix} 0.2 & 0.3 & 0.3 & 0.3 \\ 0.2 & 0.4 & 0.6 & 0.7 \\ 0.2 & 0.4 & 0.6 & 0.8 \\ 0.2 & 0.2 & 0.2 & 0.2 \end{bmatrix} \end{array}$$

$$\therefore \quad R \circ R = \begin{bmatrix} 0.2 & 0.3 & 0.3 & 0.3 \\ 0.2 & 0.4 & 0.6 & 0.6 \\ 0.2 & 0.4 & 0.6 & 0.6 \\ 0.2 & 0.2 & 0.2 & 0.2 \end{bmatrix} \quad \textbf{..... Ans.}$$

Ex.25. Define union and intersection of fuzzy set also find: (i) $R \cap S$

(ii) $R \cup S$ if

R	Y_1	Y_2	Y_3	Y_4
X_1	0.3	0.2	1	0
X_2	0.8	1	0	0.2
X_3	0.5	0.4	0.4	0

S	Y_1	Y_2	Y_3	Y_4
X_1	0.3	0	0.7	0
X_2	0.1	0.8	1	1
X_3	0.6	0.9	0.3	0.1

Soln. For definition see introduction section

(i) We have $\mu_{R\cap S} = \min\{\mu_R, \mu_S\}$

From table, we have

$$R = \left\{ \frac{0.3}{(x_1, y_1)}, \frac{0.2}{(x_1, y_2)}, \frac{1}{(x_1, y_3)}, \frac{0}{(x_1, y_4)}, \frac{0.8}{(x_2, y_1)}, \frac{1}{(x_2, y_2)}, \right.$$

$$\left. \frac{0}{(x_2, y_3)}, \frac{0.2}{(x_2, y_4)}, \frac{0.5}{(x_3, y_1)}, \frac{0.4}{(x_3, y_2)}, \frac{0.4}{(x_3, y_3)}, \frac{0}{(x_3, y_4)} \right\}$$

and $$S = \left\{ \frac{0.3}{(x_1, y_1)}, \frac{0}{(x_1, y_2)}, \frac{0.7}{(x_1, y_3)}, \frac{0}{(x_1, y_4)}, \frac{0.1}{(x_2, y_1)}, \frac{0.6}{(x_3, y_1)}, \right.$$

$$\left. \frac{0.9}{(x_3, y_2)}, \frac{0.8}{(x_2, y_2)}, \frac{1}{(x_2, y_3)}, \frac{1}{(x_2, y_4)}, \frac{0.3}{(x_3, y_3)}, \frac{0.1}{(x_3, y_4)} \right\}$$

Now,

$$\therefore R \cap S = \left\{ \frac{0.3}{(x_1, y_1)}, \frac{0}{(x_1, y_2)}, \frac{0.7}{(x_1, y_3)}, \frac{0}{(x_1, y_4)}, \frac{0.1}{(x_2, y_1)}, \right.$$

$$\frac{0.8}{(x_2, y_2)}, \frac{0}{(x_2, y_3)}, \frac{0.2}{(x_2, y_4)}, \frac{0.5}{(x_3, y_1)}, \frac{0.4}{(x_3, y_2)},$$

$$\left. \frac{0.3}{(x_3, y_3)}, \frac{0}{(x_3, y_4)} \right\}$$

$R \cap S$	Y_1	Y_2	Y_3	Y_4
X_1	0.3	0	0.7	0
X_2	0.1	0.8	0	0.2
X_3	0.5	0.4	0.3	0

Now, $(R \cup S) = \max(\mu_R, \mu_S)$

$$\therefore R \cup S = \left\{ \frac{0.3}{(x_1, y_1)}, \frac{0}{(x_1, y_2)}, \frac{1}{(x_1, y_3)}, \frac{0}{(x_1, y_4)}, \frac{0.8}{(x_2, y_1)}, \frac{1}{(x_2, y_2)}, \right.$$

$$\left. \frac{1}{(x_2, y_3)}, \frac{1}{(x_2, y_4)}, \frac{0.6}{(x_3, y_1)}, \frac{0.9}{(x_3, y_2)}, \frac{0.4}{(x_3, y_3)}, \frac{0.1}{(x_3, y_4)} \right\}$$

$R \cup S$	Y_1	Y_2	Y_3	Y_4
X_1	0.3	0.2	1	0
X_2	0.8	1	1	1
X_3	0.6	0.9	0.4	0.1

.....Ans.

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Ex.26. If R and S are fuzzy relations given by

$$R = \begin{array}{c} \\ x_1 \\ x_2 \\ x_3 \end{array} \begin{array}{c} \begin{array}{cc} y_1 & y_2 \end{array} \\ \begin{bmatrix} 0.5 & 0.1 \\ 0.2 & 0.9 \\ 0.8 & 0.6 \end{bmatrix} \end{array} \quad \text{and} \quad S = \begin{array}{c} \\ x_1 \\ x_2 \\ x_3 \end{array} \begin{array}{c} \begin{array}{cc} y_1 & y_2 \end{array} \\ \begin{bmatrix} 0.6 & 0.5 \\ 0.4 & 0.8 \\ 0.7 & 0.9 \end{bmatrix} \end{array}$$

then find $R \cup S$, $R \cap S$ and $\bar{R}$ (RS → 97Pg)

Soln. We know that $\mu_{R \cup S} = \text{Max}\{\mu_R, \mu_S\}$ and

$\mu_{R \cap S} = \min\{\mu_R, \mu_S\}$, $\mu_{\bar{R}} = 1 - \mu_R$

Now, from matrix table

$$R = \left\{ \frac{0.5}{(x_1, y_1)}, \frac{0.1}{(x_1, y_2)}, \frac{0.2}{(x_2, y_1)}, \frac{0.9}{(x_2, y_2)}, \frac{0.8}{(x_3, y_1)}, \frac{0.6}{(x_3, y_2)} \right\}$$

$$S = \left\{ \frac{0.6}{(x_1, y_1)}, \frac{0.5}{(x_1, y_2)}, \frac{0.4}{(x_2, y_1)}, \frac{0.8}{(x_2, y_2)}, \frac{0.7}{(x_3, y_1)}, \frac{0.9}{(x_3, y_2)} \right\}$$

Now,

$$R \cap S = \left\{ \frac{0.5}{(x_1, y_1)}, \frac{0.1}{(x_1, y_2)}, \frac{0.2}{(x_2, y_1)}, \frac{0.8}{(x_2, y_2)}, \frac{0.7}{(x_3, y_1)}, \frac{0.6}{(x_3, y_2)} \right\}$$

$\therefore$

$R \cap S$	y_1	y_2
x_1	0.5	0.1
x_2	0.2	0.8
x_3	0.7	0.6

Now,

$$R \cup S = \left\{ \frac{0.6}{(x_1, y_1)}, \frac{0.5}{(x_1, y_2)}, \frac{0.4}{(x_2, y_1)}, \frac{0.9}{(x_2, y_2)}, \frac{0.8}{(x_3, y_1)}, \frac{0.9}{(x_3, y_2)} \right\}$$

$\therefore$

$R \cup S$	y_1	y_2
x_1	0.6	0.5
x_2	0.4	0.9
x_3	0.8	0.9

and

$$\overline{R} = \left\{ \frac{0.5}{(x_1, y_1)}, \frac{0.9}{(x_1, y_2)}, \frac{0.8}{(x_2, y_1)}, \frac{0.1}{(x_2, y_2)}, \frac{0.2}{(x_3, y_1)}, \frac{0.4}{(x_3, y_2)} \right\}$$

$$\therefore \quad \overline{R} = \begin{array}{c} \\ x_1 \\ x_2 \\ x_3 \end{array} \begin{array}{c} \begin{array}{cc} y_1 & y_2 \end{array} \\ \begin{bmatrix} 0.5 & 0.9 \\ 0.8 & 0.1 \\ 0.2 & 0.4 \end{bmatrix} \end{array} \quad \text{..... Ans.}$$

Ex.27. If $A = \frac{0.3}{30} + \frac{0.7}{60} + \frac{1}{100}$, $B = \frac{0.2}{20} + \frac{0.4}{40} + \frac{0.6}{60}$, and $C = \frac{0.3}{50} + \frac{0.6}{100}$, then find $R = A \times B$, $S = B \times C$. Also find $R \circ S$

Soln. We have $\mu_{R=A\times B} = \min\{\mu_A, \mu_B\}$

$\mu_{S=B\times C} = \min\{\mu_B, \mu_C\}$

$\therefore$

$$R = \begin{array}{c} \\ 30 \\ 60 \\ 100 \end{array} \begin{array}{c} \begin{array}{ccc} 20 & 40 & 60 \end{array} \\ \begin{bmatrix} 0.2 & 0.3 & 0.3 \\ 0.2 & 0.4 & 0.6 \\ 0.2 & 0.4 & 0.6 \end{bmatrix} \end{array} \quad \text{and} \quad S = \begin{array}{c} \\ 20 \\ 40 \\ 60 \end{array} \begin{array}{c} \begin{array}{cc} 50 & 100 \end{array} \\ \begin{bmatrix} 0.2 & 0.2 \\ 0.3 & 0.4 \\ 0.3 & 0.6 \end{bmatrix} \end{array}$$

Now

Let $T = R \circ S$

We have

$R \circ S\,(30, 50) = \text{Max}\,\{\min\{0.2, 0.2\}, \min\{0.3, 0.3\}, \min\{0.3, 0.3\}\}$

$= \text{Max}\,\{0.2, 0.3, 0.3\} = 0.3$

$R \circ S\,(30, 100) = \text{Max}\,\{\min\{0.2, 0.2\}, \min\{0.3, 0.4\}, \min\{0.3, 0.6\}\}$

$= \text{Max}\,\{0.2, 0.3, 0.3\} = 0.3$

$R \circ S\,(60, 50) = \text{Max}\,\{\min\{0.2, 0.2\}, \min\{0.4, 0.3\}, \min\{0.6, 0.3\}\}$

$= \text{Max}\,\{0.2, 0.3, 0.3\} = 0.3$

$R \circ S\,(60, 100) = \text{Max}\,\{\min\{0.2, 0.2\}, \min\{0.4, 0.4\}, \min\{0.6, 0.6\}\}$

$= \text{Max}\,\{0.2, 0.4, 0.6\} = 0.6$

$R \circ S\,(100, 50) = \text{Max}\,\{\min\{0.2, 0.2\}, \min\{0.4, 0.3\}, \min\{0.6, 0.3\}\}$

$= \text{Max}\,\{0.2, 0.3, 0.3\} = 0.3$

$R \circ S\,(100, 100) = \text{Max}\,\{\min\{0.2, 0.2\}, \min\{0.4, 0.4\}, \min\{0.6, 0.6\}\}$

$= \text{Max}\,\{0.2, 0.4, 0.6\} = 0.6$

$$\therefore\ T = R \circ S = \begin{array}{c} \\ 30 \\ 60 \\ 100 \end{array} \begin{array}{c} \begin{array}{cc} 50 & 100 \end{array} \\ \begin{bmatrix} 0.3 & 0.3 \\ 0.3 & 0.6 \\ 0.3 & 0.6 \end{bmatrix} \end{array}$$ .....Ans.

Ex.28. If

R	Y_1	Y_2	Y_3	Y_4
X_1	0.4	0.5	0.1	0
X_2	0.8	0.9	1.0	0
X_3	0.2	0.1	0.5	0.6

and

S	Y_1	Y_2	Y_3	Y_4
X_1	0.5	0.2	0.6	0.8
X_2	0.9	1	0.1	0.1
X_3	0.8	0.9	0.6	0.7

Then find (i) $R \cup S$ (ii) $R \cap S$ (iii) $\overline{R}$

Soln. We have

(i) $\mu_{R \cup S} = \text{Max}\{\mu_R, \mu_S\}$

$\therefore$

$R \cup S$	Y_1	Y_2	Y_3	Y_4
X_1	0.5	0.5	0.6	0.8
X_2	0.9	1	1	0.1
X_3	0.8	0.9	0.6	0.7

(ii) $\mu_{R \cap S} = \min\{\mu_R, \mu_S\}$

$\therefore$

$R \cap S$	Y_1	Y_2	Y_3	Y_4
X_1	0.4	0.2	0.1	0
X_2	0.8	0.9	0.1	0
X_3	0.2	0.1	0.5	0.6

(iii) $\mu_{\overline{R}} = 1 - \mu_R$

$\therefore$

$\overline{R}$	Y_1	Y_2	Y_3	Y_4
X_1	0.6	0.5	0.9	1
X_2	0.2	0.1	0	1
X_3	0.8	0.9	0.5	0.4

.....Ans.